

A Current-limiting Control Drive Circuit for DC Motor Start-up and Blocking

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ABSTRACT

When the DC motor starts, the motor is stationary, the induction potential on the armature is zero, and the motor current rises quickly to produce a large inrush current, which will not only cause damage to the unit equipment but also cause perturbation of the power supply. In addition, in some applications encountered in the motor blocked rotation, electrical energy is fully converted to armature and MOSFET transistor thermal energy, accelerating the aging of the armature enamel wire, reducing the service life of the motor. In order to solve the problem of excessive inrush current when starting and blocking the motor, this paper presents an H-bridge a current-limiting control drive circuit for DC motor start-up and blocking. The circuit designed in this paper adds a current detection circuit to the driving circuit, and uses the parasitic inductance of the motor armature to effectively limit the peak current of the motor through the control of a fixed shutdown time to achieve the current limiting effect. The test results show that the motor can limit the current output during starting and blocking, and the current output is good without affecting the normal operation of the motor.

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1 INTRODUCTION

DC motors are widely used in small appliances, toys, cell phones and other portable equipment systems because of their wide range of speed regulation and low cost, which creates the conditions for the application of DC motors. In order to make the DC motor work properly, it is necessary to have some special control and protection devices to avoid damage due to excessive starting current and long-term overload, and to extend the service life of the motor [1].

The starting of a DC motor is a process in which the motor is turned on from a standstill and accelerated to a stable operating speed. The whole starting process is a transition process and the

circuit equation is as follows:

$$U = e_a + i_a \sum R_a + L_a \frac{di}{dt} \quad (1)$$

U is the external voltage of the motor; e_a is the induced electric potential on the armature; $\sum R_a$ is the total resistance of the armature circuit; and L_a is the armature inductance. The current in the loop increases much faster than the motor's starting speed, so L_a can be neglected. When the motor is stationary at start-up, so that e_a is 0, the equivalent DC voltage on the motor flows through the loop resistance of the armature. If the loop resistance is small, it will result in an excessive starting current [2]. Excessive current at motor start-up will cause mechanical shock to the unit, sparks to the commutator, damage to the transistor with low overload capacity, and system power disturbance, which will affect the normal operation of other equipment. The problem of excessive current can also occur when the motor is overloaded or blocked. There are currently three methods of starting a DC motor:

1. Direct starter

This method can be used for systems with a small range of motor supply voltage variations, large armature series resistance, high commutation transistor overload capability, and low starting speed requirements[3,4].

2. Armature Series Resistance Starters

By increasing $\sum R_a$ through the series resistance, the starting and stall currents can be reduced. For systems with varying motor supply voltages, the motor starting current and stall will vary depending on the motor supply voltage, and when the motor voltage is low, the starting speed will be significantly reduced. Moreover, series resistors consume energy all the time, which seriously reduces the efficiency and effective working time of portable battery-powered systems[5,6].

3. Buck starter

Adding a constant voltage regulator to provide a lower and constant voltage for the motor to reduce the starting and stall currents, so that the voltage on the motor remains unchanged under different supply voltages and the starting current and stall current are also unchanged, but still cannot solve the problem of heat loss [7, 8].

In order to control the starting and stall currents, avoid thermal losses, and improve system efficiency, this paper presents an H-bridge DC reversing motor current limiting drive circuit. The circuit utilizes an added current detection circuit and a fixed off-time control mode to control the starting and stall currents to avoid the above-mentioned problems. Section 2 introduces general structure of DC motor drive circuit. DC motor drive circuit design in section 3. In Section 4 flow limitation principle is presented and test results

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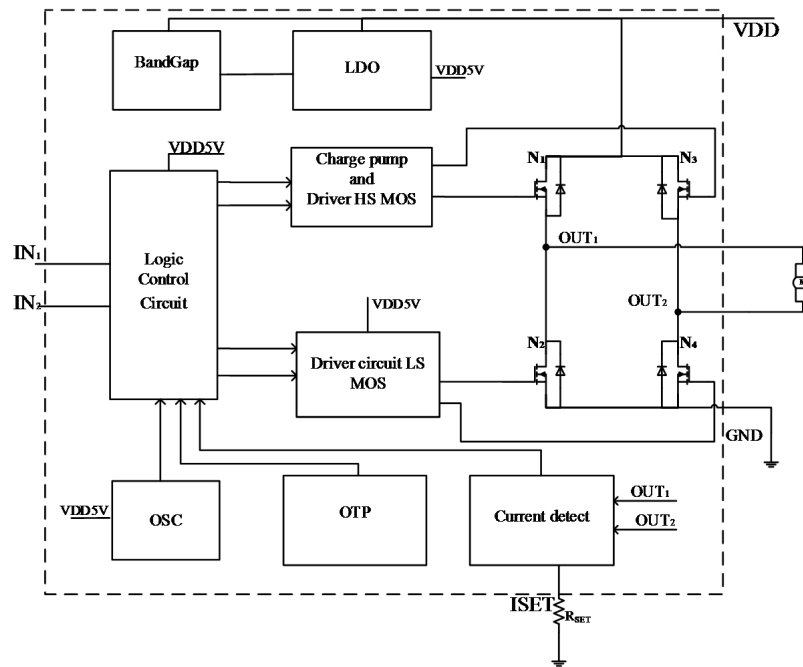


Figure 1: DC Motor Drive Circuit Block Diagram

are presented on the use of the driver chip in a toy car. Section 5 conclude.

2 GENERAL STRUCTURE OF DC MOTOR DRIVE CIRCUIT

Figure 1 shows a general block diagram of the DC motor drive circuit. As shown in Figure 1, the DC motor drive circuit mainly consists of an LDO circuit, an H-bridge drive circuit, a drive signal amplifier circuit, a current detection circuit, a fixed off-time timing circuit, a charge pump circuit, a control logic circuit, and an overheat and under-voltage protection circuit. The peripheral interface of the motor drive circuit is simple. The main interfaces are IN_1 and IN_2 signals to control the running direction of the motor, ISET for current limit setting, VDD for motor power, and OUT_1 and OUT_2 for DC motor interface.

In the DC motor drive circuit design, the maximum motor voltage is 16V, so an internal LDO circuit is required to generate constant 5V to supply the 5V to the low-voltage circuit. The current detect low voltage module provides the reference voltage. Charge pump and Driver HS MOS module drives the high side NMOS transistors N_1 , N_3 . Driver circuit LS MOS is responsible for driving the low side NMOS transistors N_2 , N_4 . The current detect module is responsible for detecting the current flowing through the power tube, generating an overcurrent protection signal, and then transmitting the overcurrent signal to the Logic Control Circuit module for current limit control. The OSC is an oscillator module that provides the clock signals for the Logic Control circuit and the charge pump module circuit.

3 DC MOTOR DRIVE CIRCUIT DESIGN

3.1 H-Bridge Drive Circuit Design

In the DC motor control circuit design, this paper designs the H-bridge circuit as the power drive circuit of the driver [9]. Power MOSFET transistors have the characteristics of large input impedance and fast switching speed to meet the needs of high-speed switching [10], so the power MOSFET transistors are commonly used as the bridge arm of the H-bridge. Four power MOSFET transistors are required in the H-bridge circuit. Due to the high carrier mobility of N-type MOSFET transistors, the conduction circuit per unit area is small, and N-type MOSFET transistors are selected for both the upper and lower bridge arms due to the cost of circuit fabrication. As shown in Figure 1, among the four transistors, transistors N_1 and N_4 are one group, transistors N_2 and N_3 are another group, and when one group of transistors is turned on, the other group of transistors must remain off. To prevent the upper and lower two groups of transistors to prevent collusion between the two groups and burned out, it is necessary to increase the delay between the upper and lower two groups of control signals to provide a long enough dead time, in addition to the addition of logic control signals to limit the same effect can also be achieved.

3.2 Charge Pump Circuit Design

For H-bridge lower side transistors N_2 and N_4 , the transistors can be turned on and off directly with a 5V drive circuit, and for H-bridge high side transistors with NMOS transistors, the gate source voltage needs to reach 5V for NMOS transistors N_1 and N_3 to be fully turned on, and for transistor N_1 , for example, when transistors N_1 and N_4 are turned on at the same time, the OUT_1 terminal is

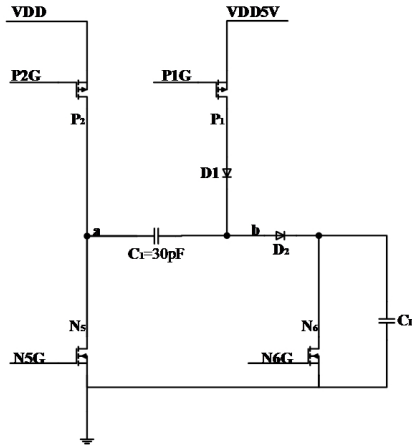


Figure 2: Charge Pump Schematic

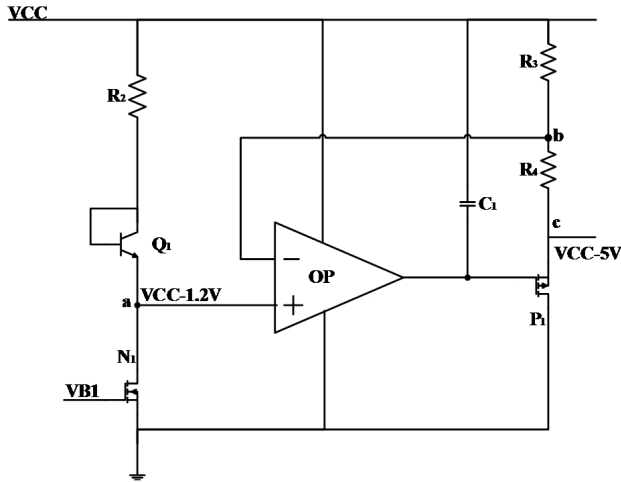


Figure 3: VCC-5V LDO Driver Circuit Schematic Diagram

transistor N_1 . The source potential of the transistor needs to be close to VCC, so the gate potential of the transistor M_1 needs to be greater than the VCC voltage. In order to achieve the above function, this paper designs the charge pump circuit. The charge pump schematic is shown in Figure 2

As shown in Figure 2, P_1 is the low-voltage PMOS transistor, P_2 is the high-voltage PMOS transistor, N_1 and N_2 are the high-voltage NMOS transistors, $P1G$, $P2G$, $N5G$, $N6G$ are the gate drive signals of the four MOSFET transistors, C_1 is the internal integrated capacitor, C_L equivalent switch gate capacitor, $P1G$, $P2G$, $N5G$ provide square wave signal. The charge pump circuit works as follows: when transistors P_2 and N_2 are off, transistors P_1 and N_1 are on, VDD charges capacitor C_1 , and if the forward voltage drop of the diode is ignored, $V_{ab} = -5V$. Then we put transistors P_1 , N_1 and N_2 in the off state, the transistor P_2 is on, the voltage at a terminal is equal to VCC, and the charge on capacitor C_1 is charged by the $V_{ab} = -5V$. As the diode D_1 moves across the C_L , the charge moving across the C_L will remain

stable due to the unidirectional conductivity of the diode. After a certain period of operation, the voltage across the C_L will be close to $VCC + 5V$ (ignoring the diode's forward voltage drop), and the H-bridge high side NMOS transistors N_1 , N_3 will be fully turned on. When it is necessary to turn off high side NMOS transistors N_1 and N_3 in the H-bridge circuit, only transistors P_1 and P_2 need to be kept off; transistors N_5 and N_6 need to be kept on.

From the above, it can be seen that the high-side MOS transistors in the H-bridge circuit take a certain amount of time to turn on, and the amount of time depends on the capacitance C_1 , C_L , and the oscillation frequency of the charge-pump circuit. In this design, the oscillation frequency is 1MHz, capacitance C_1 is selected at 30pF, and the high-side transistor open time is 100us. To ensure that the high side MOS transistor is turned on first and then the corresponding lower MOS transistor is turned on, the lower side of the MOS transistor control signal needs to be delayed 200us. The delay time of 200us is negligible for most motor drive applications.

3.3 VCC-5V LDO Driver Circuit Design

In Figure 2, the gate control signals of transistors P_1 and N_1 need to be high to 5V, so the drive circuit power can be completed by 5V LDO driver circuit, while transistor P_2 needs its gate voltage to be lower than the supply voltage of 5V, so we need to design VCC-5V LDO driver circuit.

Figure 3 shows the schematic of the VCC-5V LDO driver circuit. First, the Bandgap module generates a bias current I_{bias} proportional to the absolute temperature. I_{bias} can be expressed as:

$$I_{bias} = \frac{V_T \ln 8}{R_1} \quad (2)$$

Where V_T is the voltage equivalent of temperature, 26mV at room temperature, and R_1 is the resistance of the bandgap reference circuit; it is mirrored by the mirror circuit into transistor N_1 . Then the voltage generated at point a is given by:

$$V_a = VCC - \left(\frac{V_T \ln 8}{R_1} \times R_2 + V_{BE1} \right) \quad (3)$$

According to the design principle of the bandgap reference, a bandgap reference voltage of about 1.2V can be obtained by taking reasonable values for R_1 and R_2 . Then the voltage generated at point a can be expressed as:

$$V_a = VCC - 1.2 \quad (4)$$

Since op is the ideal op-amp, the voltage at point b is equal to the voltage at point a. The voltage at point c is calculated as follows:

$$V_c = VCC - \frac{(VCC - V_b) \times (R_2 + R_4)}{R_2} \quad (5)$$

In this paper, the ratio of R_4 and R_3 is designed to be: 4.167, and the c-point voltage VCC-5V provides the equivalent ground potential for the driver stage circuit. Where transistor P_1 is the source follower, which has a strong current driving capability. Because the drive stage speed is faster, transient current is larger, in order to reduce the VCC-5V ripple, the transistor P_1 width to length ratio of 200um/0.5um, while increasing the capacitance C_1 can also reduce the output ripple voltage.

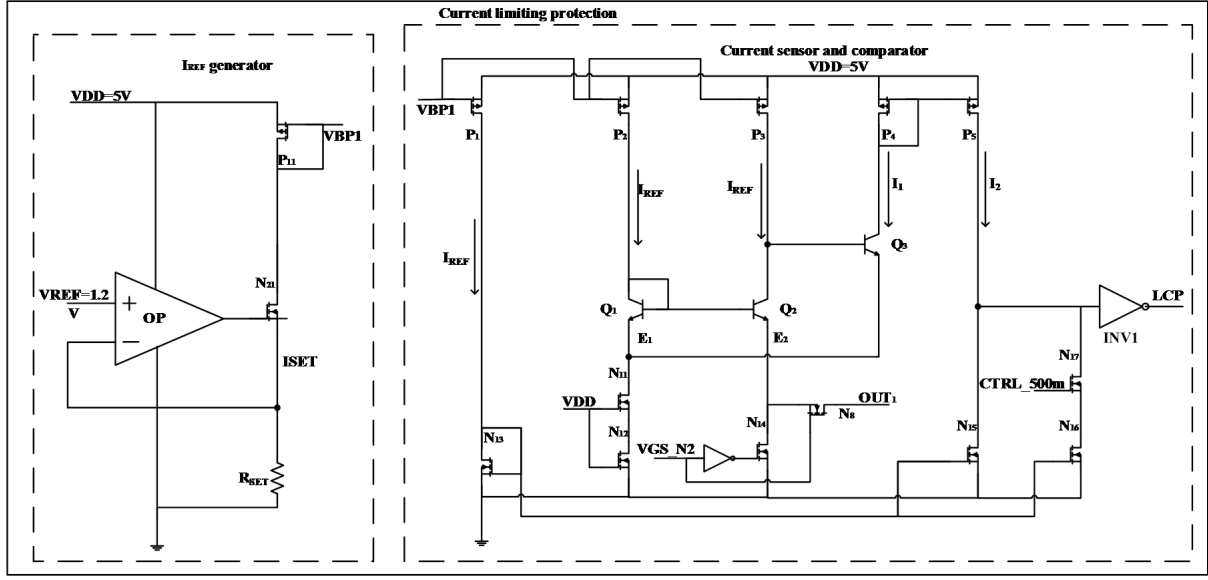


Figure 4: Current Detection Circuit Schematic

3.4 Current Detection Circuit Design

In order to limit the current of a DC motor, we need to detect the motor current in real time. DC motors output current through power switches, so detecting the current in power switches transistors N_1 , N_2 , N_3 and N_4 is the same as detecting the motor current. The high side transistors N_1 and N_3 are driven by a charge pump and the sampling circuit design is complex. In order to simplify the detection circuit design, this paper chooses to detect the currents of low-side transistors N_2 and N_4 . The current detection circuit schematic of OUT_1 low-side transistor N_2 , for example, is shown in Figure 4

Transistors P_1 , P_2 and P_3 are current sources, the current is I_{REF} . N_{11} and N_{12} are high-voltage MOS transistors of the same type as power tubes, and their transistor aspect ratio size is small. Transistors P_2 and P_3 form amplifiers with Q_1 , Q_2 , Q_3 , and Q_4 to equalize the voltage at the two points E_1 and E_2 . Transistors N_{13} , N_{15} , N_{16} form a current mirror with a mirror ratio of 2: m: m. The motor current mirrored by transistor P_5 and the reference current mirrored by transistor N_{15} form a current comparator that outputs a current comparison digital signal to the control module after passing through INV1.

The signal VGS_N2 is high voltage when power transistor N_2 is on, transistor N_8 is on, transistor N_{14} is off, and the OUT_1 voltage is transmitted to point E_2 , and the voltage at point E_1 is equal to the OUT_1 voltage; the on-resistance of power transistor N_2 is R_{dson} , and the current flowing through the motor is I_M , then the OUT_1 voltage is given by:

$$V_{OUT1} = I_M R_{dson} \quad (6)$$

When transistors N_{11} and N_{12} are conducting, R_{N12} is a series resistor of transistors N_{11} - N_{12} , and the reference current I_{REF} is much smaller than the motor current I_M , the sampling current I_1 is

given by:

$$I_1 = \frac{(I_{REF} + I_M) R_{dson}}{R_{N12}} - I_{REF} \approx \frac{I_M \times R_{dson}}{R_{N12}} - I_{REF} \quad (7)$$

When the current I_2 sampled by transistor P_5 is compared with the reference current I_{REF} , the output signal of inverter INV1 is reversed, and the corresponding currents of transistors P_5 and N_{15} are approximately equal, both mI_{REF} , and the corresponding motor currents are given by:

$$I_M = \frac{(m+1) \times I_{REF} \times R_{N12}}{R_{dson}} \quad (8)$$

In the equation, R_{N12} and R_{dson} are both on-resistors of the same type of MOSFET, which are consistently affected by the process angle and temperature, so the ratio of width to length remains constant, reducing the dispersion of the set current limit value. R_{N12} and R_{dson} in this design are 20kohm, 100mohm, $m=1.5$.

When the power transistor N_2 is off, the signal VGS_N2 is low, transistor N_8 is off, transistor N_{14} is on, the E point voltage is 0, and the module output is constant 0, maintaining the non-current state.

The bias current I_{REF} directly determines the current limit value of the motor operation. The source terminal of transistor N_{21} is the output port of the chip, and the external resistor R_{SET} can set different current-limiting thresholds. The current I_{REF} flowing through the operational amplifier feedback $P1$ is given by:

$$I_{REF} = V_{REF} / R_{SET} \quad (9)$$

Substitute equation 9 into equation 8, I_M can be expressed as:

$$I_M = \frac{V_{REF} \times (m+1) \times R_{N12}}{R_{SET} \times R_{dson}} \quad (10)$$

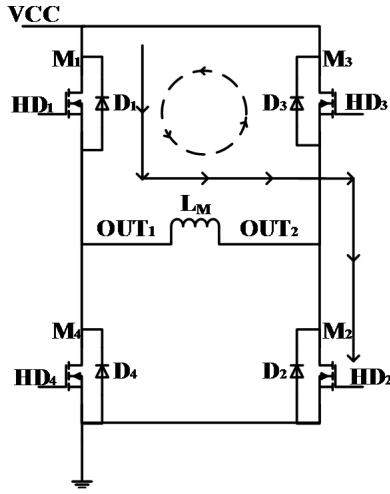


Figure 5: Schematic diagram of motor current limiting operation

4 FLOW LIMITATION PRINCIPLE AND APPLICATION TESTING

4.1 Current limiting circuit principle of operation

The schematic diagram of the current limiting circuit in operation is shown in Figure 5. As shown in figure 5, when transistors M_1 and M_2 are conducting, OUT_1 is a high voltage, OUT_2 is low voltage, motor current flows from OUT_1 to OUT_2 , as shown in the figure shown in the solid line, when the detection circuit detects the current flowing through M_2 is equal to the current limit set value, the detection circuit will pass the signal to the control logic unit, logic control unit output control signal to the low side drive MOS circuit, the low-side drive MOS circuit keeps the M_2 transistor in a disconnected state. Due to the continuity of the inductive current, the current flows through diode D_3 back to the M_1 drain, and flows through diode D_3 , the voltage at both ends of the motor is negative, the current will decrease, after a short period of time, the current decreases to below the detection circuit limit value, the M_2 transistor turns on again, the current continues to flow through M_1 and M_2 , and the output current increases, when the output current increases to the current limit value, M_2 is turned off again, and repeat until the control signal IN_1 is turned off.

In Figure 6, I_M is the motor current waveform. In order to increase the speed of the motor when starting or steering, the current limit value is 1.5 times of I_{SET} in 500ms after the motor starts, and I_{SET} is the current limit value of the output current when the DC motor is working normally. The conversion control of the DC motor output current limit value is achieved by timing the IN_1 or IN_2 signal with a clock signal. When the clock signal IN_1 or IN_2 goes high, transistor N_{17} in Figure 4 opens and current is limited to $1.5 I_{SET}$. After 500 ms, transistor N_{17} shuts down and current limit of output current is I_{SET} . When the clock signal IN_1 or IN_2 goes low, the timing circuit is reset until IN_1 or IN_2 goes high, then the timing circuit resets the timing and controls the state of transistor N_{17} . Since the timing

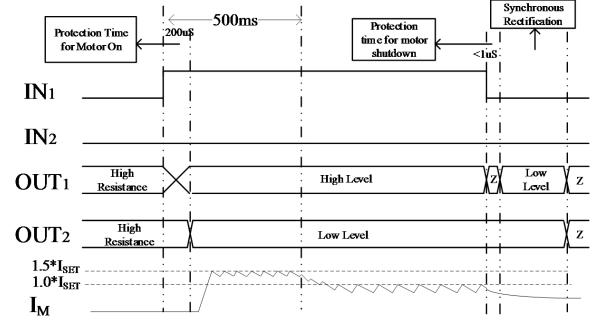


Figure 6: Ideal waveform for motor current limiting operation

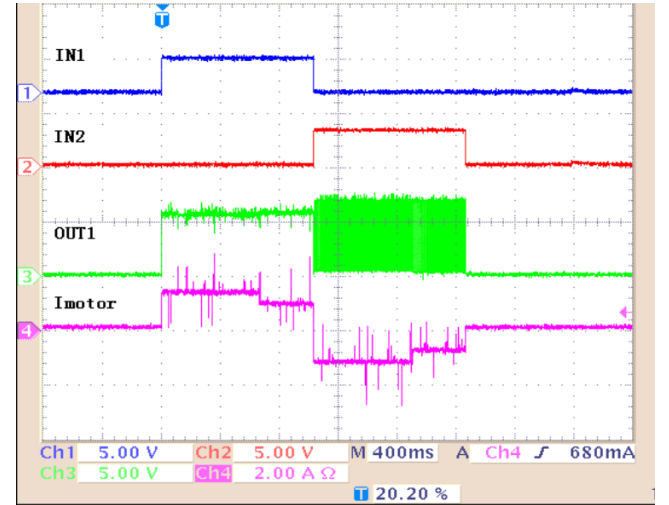


Figure 7: Application testing results

of the timing circuit is not required to be very accurate in actual application, the circuit does not correct the clock signal and the deviation is large.

4.2 Application testing results

To test the current-limiting effect of starting and blocking the DC motor, we applied the designed circuit to the front and rear wheels of the toy car. As shown in Figure 7, 1 and 2 channels are motor control signals IN_1 and IN_2 , 3 channels are drive circuit outputs OUT_1 , and 4 channels are motor current waveforms. When the IN_1 clock signal rises to 5V, IN_2 keeps 0V, the DC motor starts working, OUT_1 outputs a voltage signal of about 5V, the DC motor current output is limited to 1.5A, after 500ms the output current is limited to 1A, the control signal IN_2 rises 5V, IN_1 falls to 0V, the DC motor output current is reversed, then the output current is limited to -1.5A After 500ms, the output current is limited to 1A constant current. After switching the motor steering, the output current also has a current limiting effect, but the currents are reversed. The burr in the motor current waveform is an interference from the test.

5 CONCLUSION

In this paper, the proposed H-bridge DC commutator motor current-limiting drive circuit uses a current detection circuit and a fixed shutdown time control mode to control the starting and blocking currents of the DC motor, which has a good current-limiting effect. The proposed DC motor drive circuit is applied to the front and rear wheels of a toy car to test its starting and blocking effects. The results show that the current limiting scheme of the DC brushed motor drive can achieve the current limiting function when the motor starts and clogs up, which can limit the inrush current and reduce the heat generation of the enamel wire, and prolong the service life of the motor. Of course, this design is using the peak current detection method, for the case of large motor parasitic inductance, the ripple current of the motor parasitic inductance is small, and the error of the current limit value can be controlled within a comparable range, but for the case of small motor parasitic inductance, the ripple current of the motor parasitic inductance is large, and the current limit value will be much smaller than the peak current, and the current limit effect becomes worse, which needs to be improved.

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